

Christopher Skora

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EDUCATION

University of Waterloo

Waterloo, ON

BASc, Nanotechnology Engineering (Co-op)

Sept. 2023 – Expected Apr. 2028

- **Selected Coursework:** Microfabrication & Thin Films, Nanoprobng & Lithography, Electronic Circuits, Structures & Properties of Nanomaterials

PROFESSIONAL EXPERIENCE

RAMPS Detector & DAQ Researcher

Jan. 2026 – Apr. 2026

SNOLAB

Lively, ON

- Built and commissioned SNOLAB's first underground HPGe-SiPM coincidence detector through hands-on assembly and integration of a dual-coldhead HPGe, custom SiPM mounting fixture, and DAQ system.
- Reduced β^- background by $\sim 99\%$ by developing a Python pipeline to process raw binary DAQ output and isolate detector waveforms, enabling HPGe energy calibration with $\sim 3\%$ resolution.
- Debugged hardware-firmware integration issues on a pre-release CAEN DT2751 digitizer, resolving acquisition faults in collaboration with CAEN engineering.
- Designed and deployed a VNC-based remote DAQ and monitoring system, enabling 30+ hours of unattended operation and collection of 20+ TB of waveform data during the first science run.

SNO+ QA & Cleanliness Assistant

Sept. 2025 – Dec. 2025

SNOLAB

Lively, ON

- Measured wavelength-shifter emission spectra and timing across multiple laser excitation wavelengths using the SNO+ laserball calibration system, combining ROOT and Python for test-data analysis.
- Performed process quality assurance on LAB/PPO/Bis-MSB scintillator samples using UV-Vis spectrophotometry, turbidity, and density analysis across sampling ports throughout the SNO+ plant.
- Executed monthly quality-control inspections across a 5,000 m² Class 100 underground laboratory using particulate counting and swipe testing; documented and reported contamination status to collaborating groups.

PROJECTS & LAB EXPERIENCE

MIS Device Fabrication

May 2026 – Aug. 2026

University of Waterloo

- Executed an end-to-end cleanroom fabrication process for MIS capacitors on p-type Si, integrating RCA cleaning, PECVD SiNx deposition, photolithography, RIE, Al sputtering, and PAN wet etching.
- Performed optical defect inspection of SiNx dielectric and Al contact structures, identifying patterning and contact imperfections following selective etching to inform process improvement.
- Analyzed process metrology and electrical test data using ellipsometry, DEKTAK profilometry, four-point probe, and C-V/I-V sweeps to quantify film thickness, uniformity, resistivity, and device performance.

EV Diagnostics & Systems Design

Oct. 2023 – Apr. 2024

UWAFTElectric Car Design Team

- Converted EV electrical schematics into PCB layouts and designed mechanical enclosures using AutoCAD and SolidWorks, supporting electrical and mechanical system integration.
- Applied statistical process control (SPC) and structured root-cause analysis to diagnose Cadillac Lyriq electrical faults and improve system performance.

TECHNICAL SKILLS

Manufacturing & Quality: Hands-on Prototyping, Cleanroom Fabrication, Process Characterization, Defect Inspection, Quality Assurance, SPC, Root-Cause Analysis

CAD & Design: SolidWorks, AutoCAD, PCB Layout, Mechanical Enclosure Design

Fabrication Processes: Photolithography, PECVD, RIE, Al Sputtering, Wet Etching, Ellipsometry, DEKTAK Profilometry

Electrical & Instrumentation: C-V/I-V, Four-Point Probe, Probe Station, HPGe, SiPM, CAEN DT2751, WaveDump2, CoMPASS

Data & Automation: Python, ROOT, MATLAB, DAQ Integration, Test-Data Analysis, Calibration, Remote Monitoring